

Signals and Systems

Chapter 1

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Signals and Systems Outline

- Size of a Signal
- Useful Signal Operations
- Classification of Signals
- Signal Models
- Classification of Systems
- System Model: Input-Output Description

Size of Signal-Energy Signal

- Signal: is a set of data or information collected over time.
- If the signal goes to zero as time goes to infinity then the signal is measured by its energy E_x :

$$E_x = \int_{-\infty}^{\infty} |x(t)|^2 dt$$

Example: Find the energy of the signal $x(t) = 3e^{-2t} \quad t \geq 0$

Size of Signal-Power Signal

If the signal is periodic or the amplitude of $x(t)$ does not $\rightarrow 0$ when $t \rightarrow \infty$ ", need to measure power P_x instead:

$$P_x = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} |x(t)|^2 dt$$

Example: Find the power of the signal $x(t) = A\cos(100t)$

Useful Signal Operations

- Time Delay
- Times Scaling
- Time Reversal

Time Delay

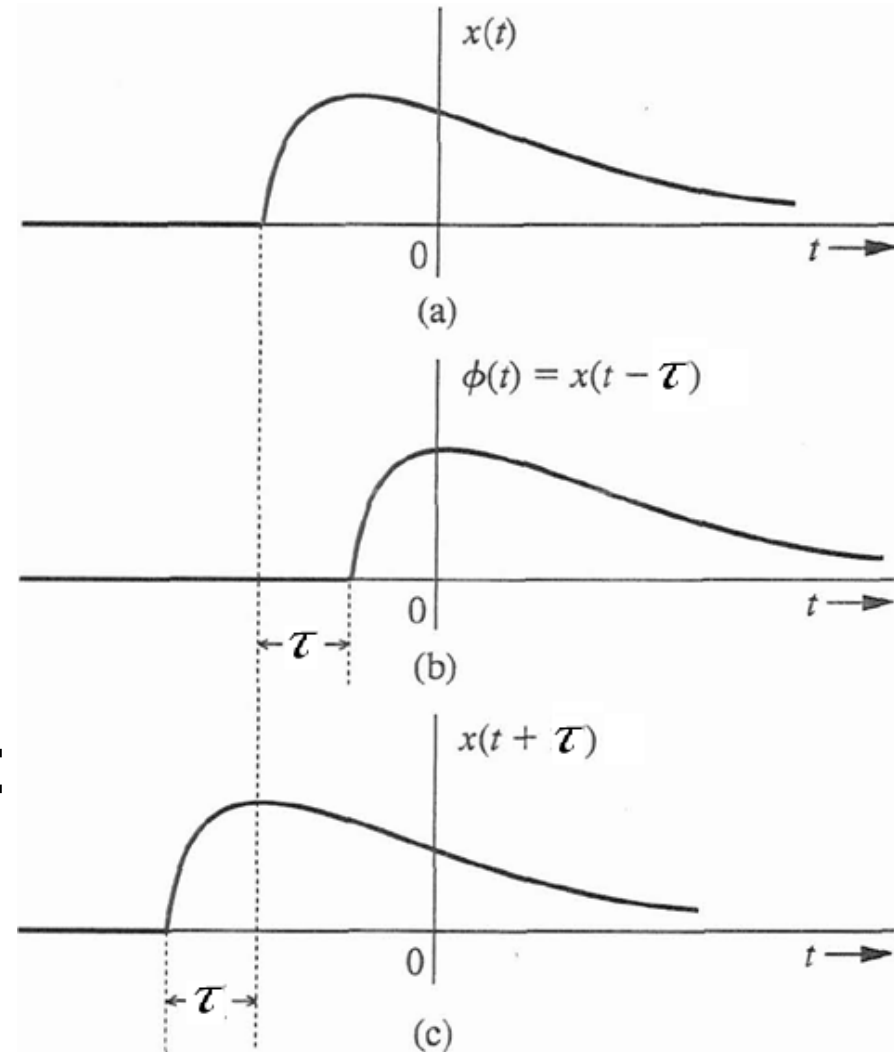
Signal $x(t)$

$x(t)$ delayed by time τ :

$$\phi(t) = x(t - \tau)$$

$x(t)$ advanced by time τ :

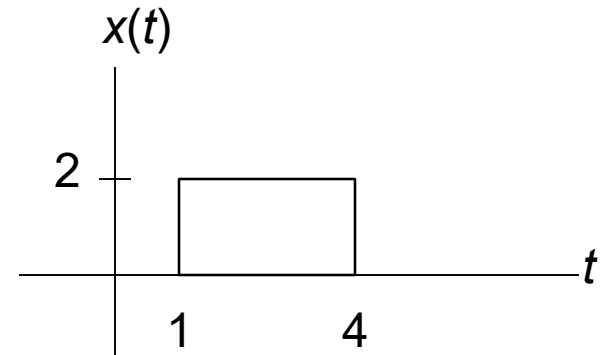
$$\phi(t) = x(t + \tau)$$



Time Delay Example

Find $x(t-2)$ and $x(t+2)$ for the signal

$$x(t) = \begin{cases} 2 & 1 \leq t \leq 4 \\ 0 & \text{elsewhere} \end{cases}$$



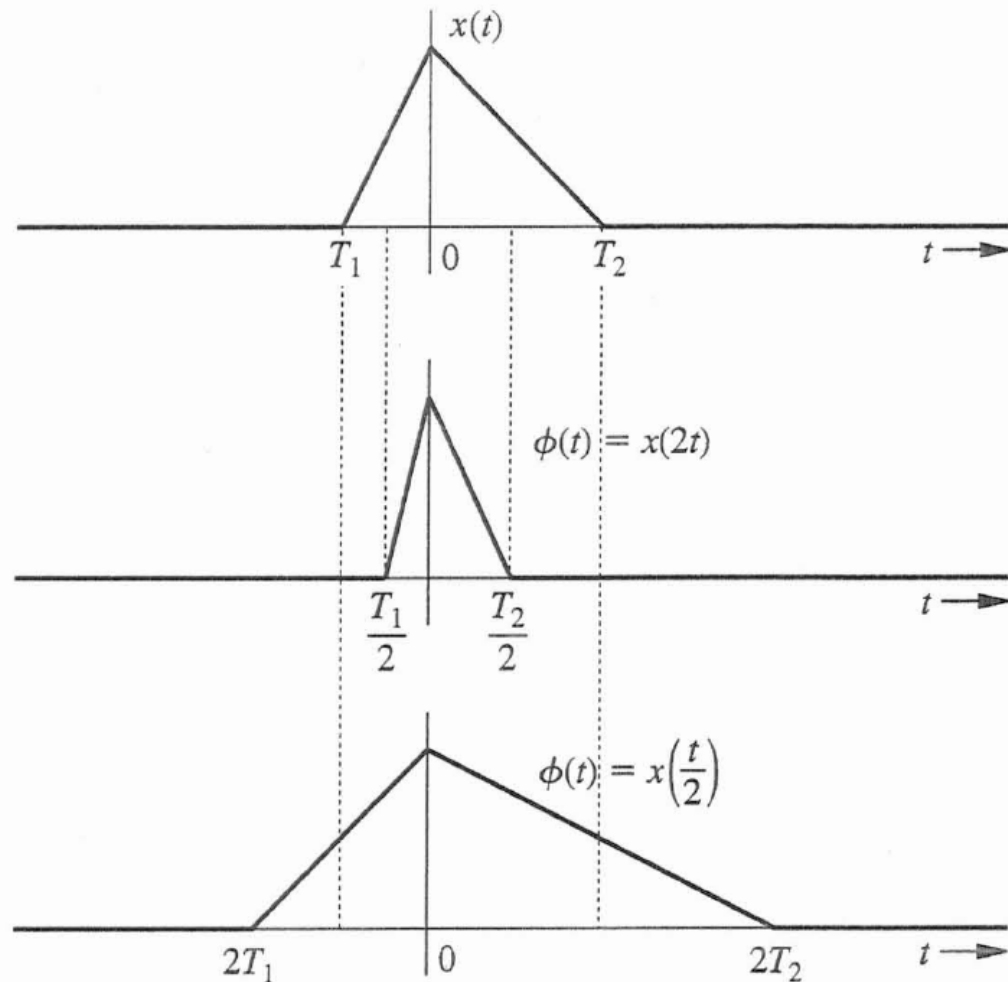
Time Scaling

$x(t)$ compressed in time
by a factor of 2:

$$\phi(t) = x(2t)$$

$x(t)$ expanded in time
(by a factor of 2):

$$\phi(t) = x(t/2)$$

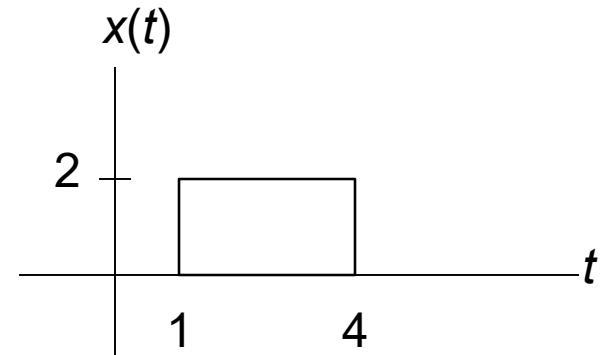


Same as recording played back at twice and half the speed respectively

Time Scaling Example

Find $x(2t)$ and $x(t/2)$ for the signal

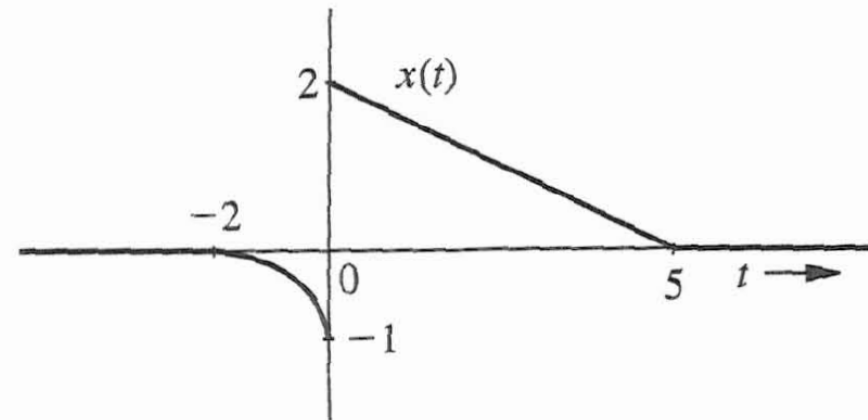
$$x(t) = \begin{cases} 2 & 1 \leq t \leq 4 \\ 0 & \text{elsewhere} \end{cases}$$



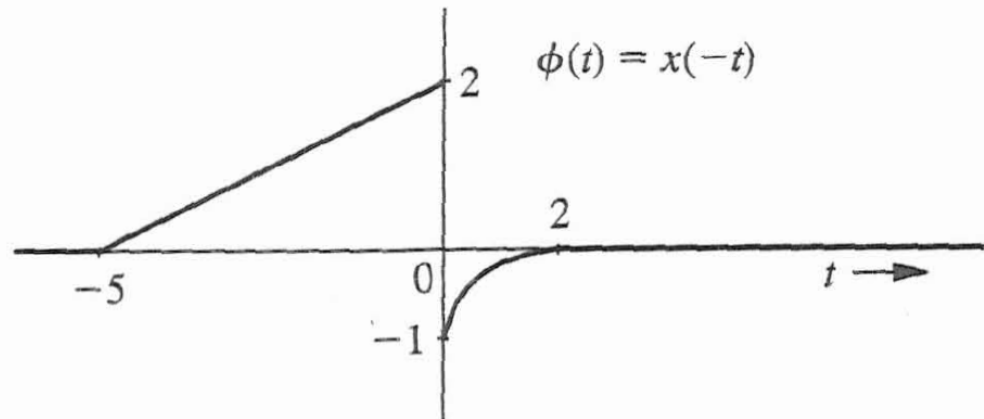
Time Reversal

Signal may be reflected about the vertical axis (i.e. time reversed):

$$\phi(t) = x(-t)$$



(a)

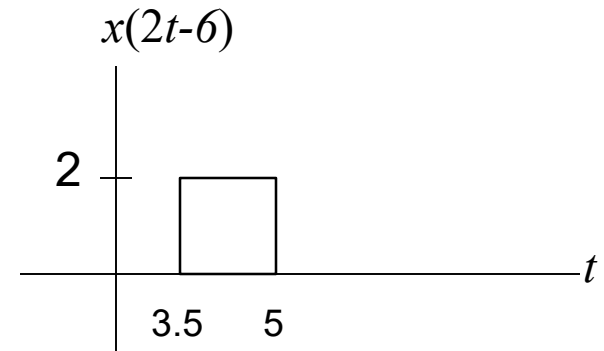
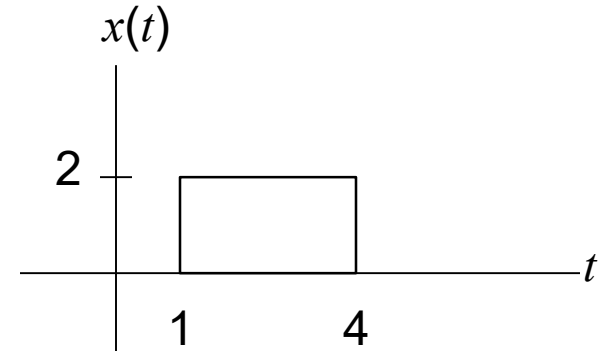


Example

Find the signal $x(2t - 6)$

can be obtained in two ways;

- Delay $x(t)$ by 6 to obtain $x(t - 6)$, and then time-compress this signal by factor 2 (replace t with $2t$) to obtain $x(2t - 6)$.
- Alternately, time-compress $x(t)$ by factor 2 to obtain $x(2t)$, then delay this signal by 3 (replace t with $t - 3 \rightarrow x(2(t-3))$, to obtain $x(2t - 6)$.



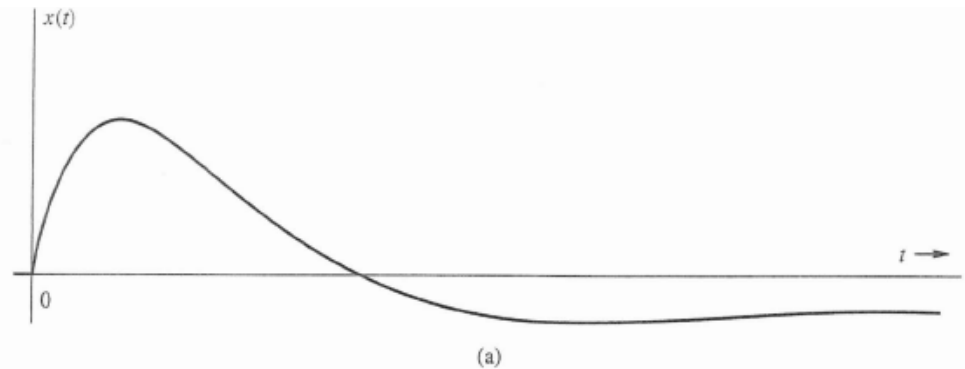
Signal Classification

Signals may be classified into:

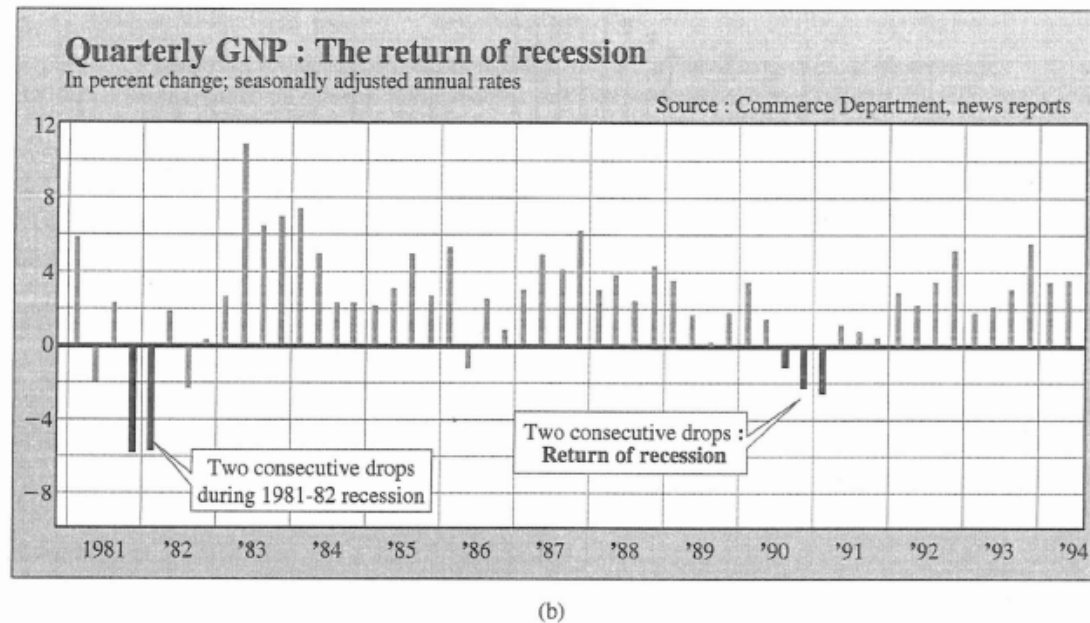
1. Continuous-time and discrete-time signals
2. Analog and digital signals
3. Periodic and aperiodic signals
4. Energy and power signals
5. Deterministic and probabilistic signals
6. Causal and non-causal
7. Even and Odd signals

Continuous vs Discrete

Continuous-time

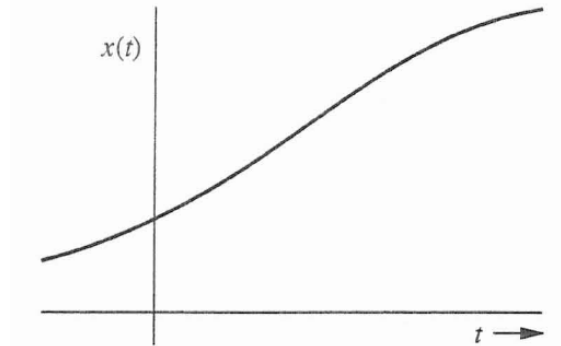


Discrete-time

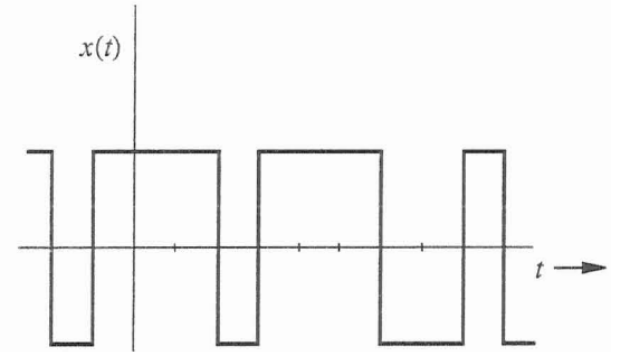


Analog vs Digital

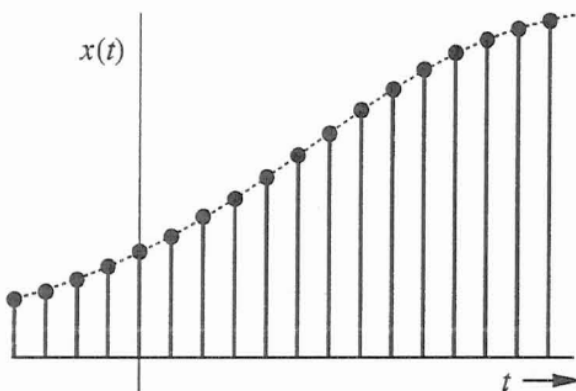
Analog, continuous



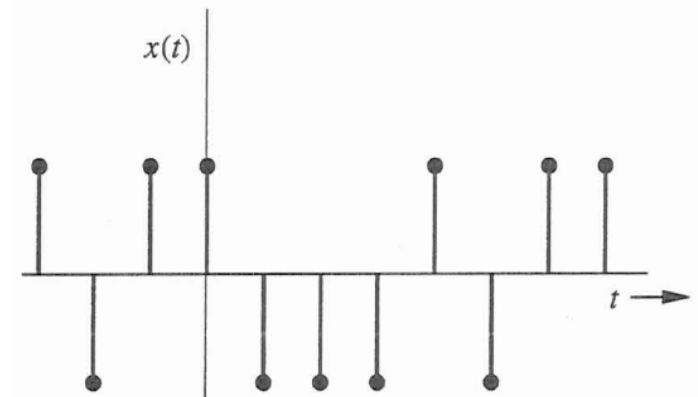
Digital, continuous



Analog, discrete



Digital, discrete

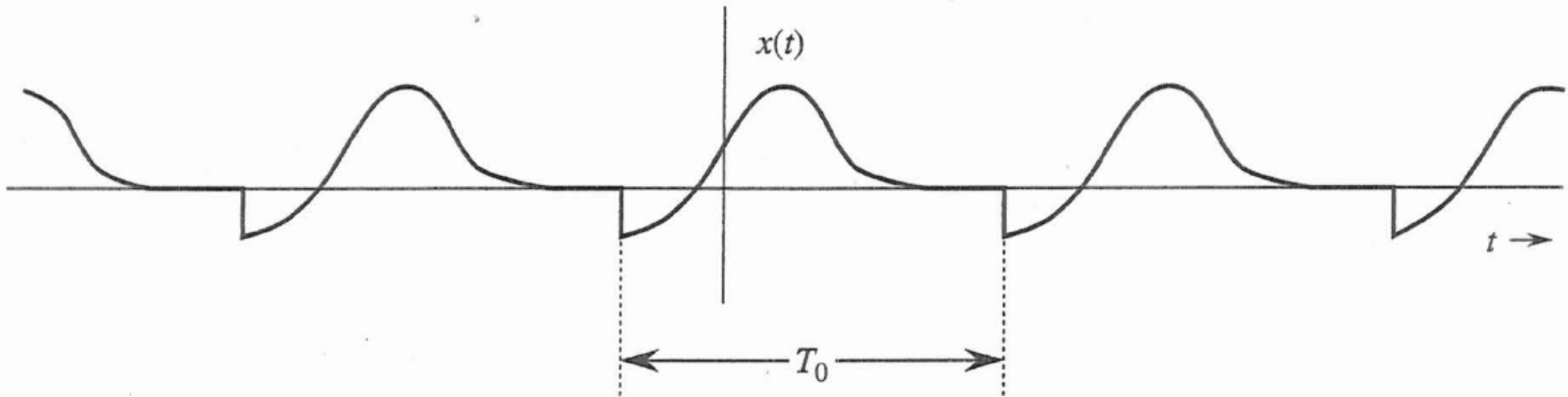


Periodic vs Aperiodic

A signal $x(t)$ is said to be periodic if for some positive constant T_0

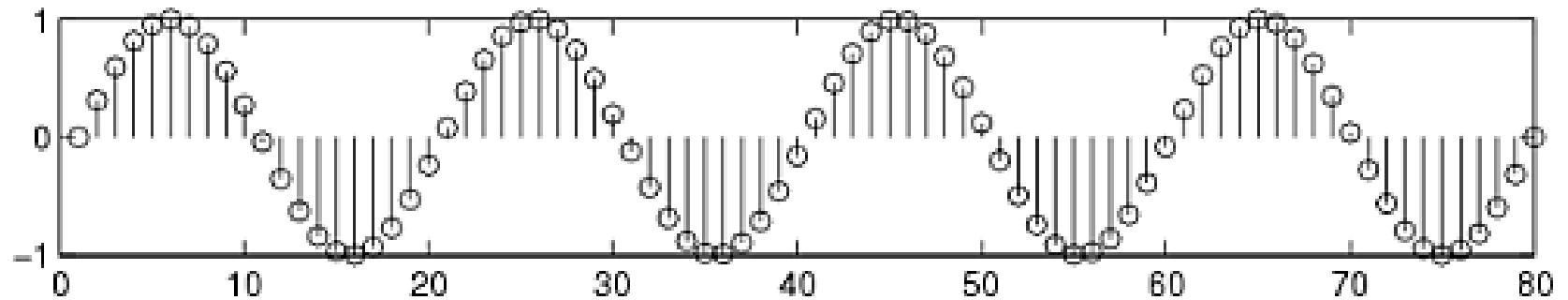
$$x(t) = x(t + T_0) \text{ for all } t$$

The smallest value of T_0 that satisfies the periodicity condition of this equation is the fundamental period of $x(t)$.

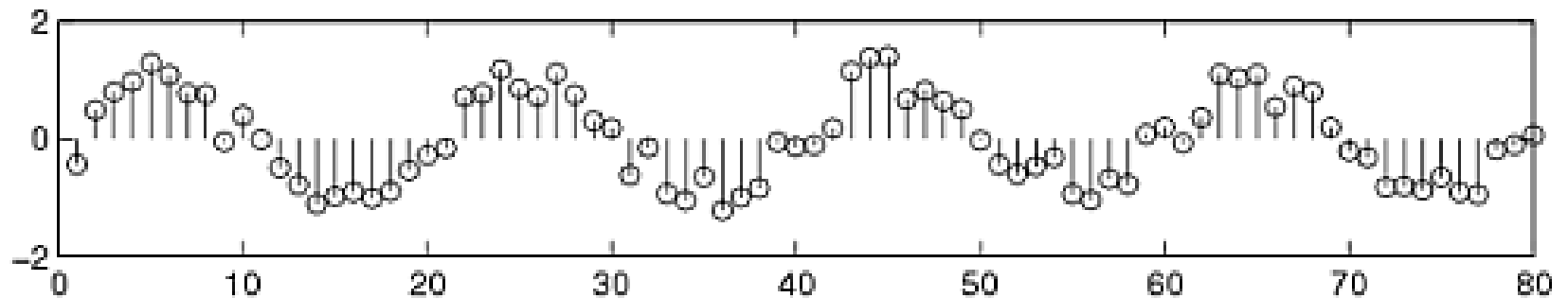


Deterministic vs Random

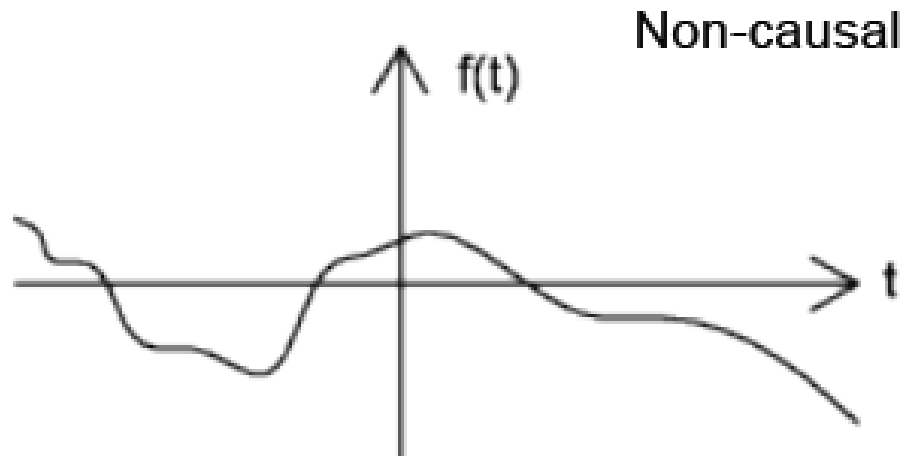
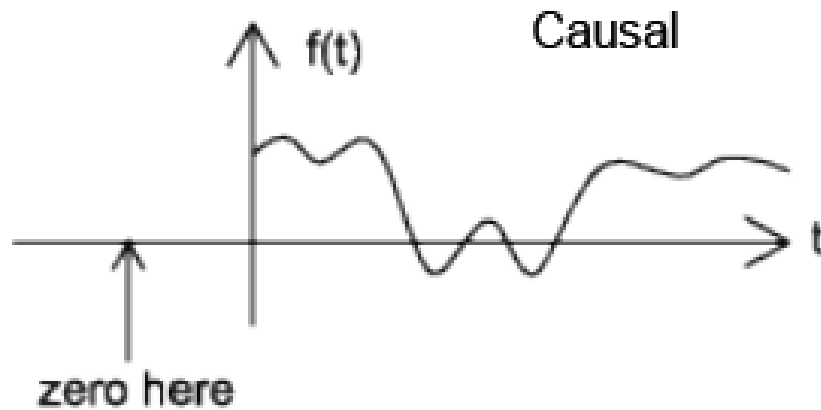
Deterministic



Random



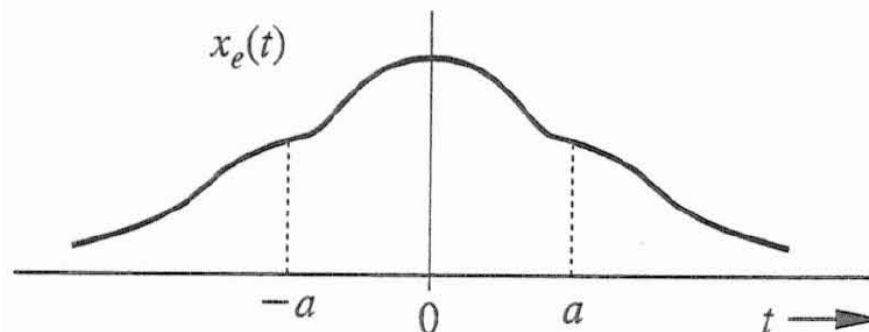
Causal vs Non-causal



Even and Odd Functions

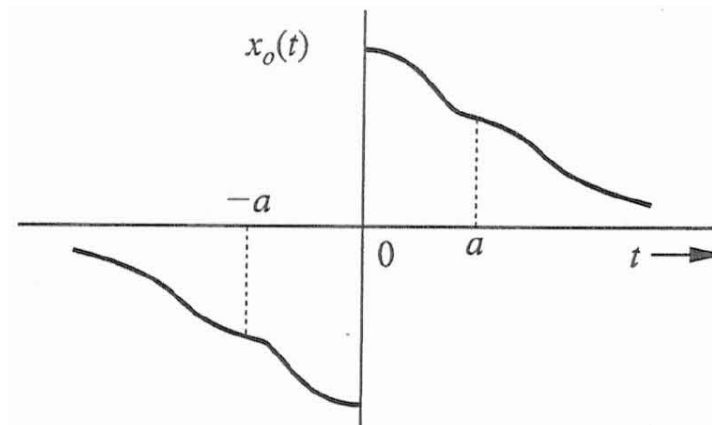
A real function $x_e(t)$ is said to be an even function of t if

$$x_e(t) = x_e(-t)$$



A real function $x_o(t)$ is said to be an odd function of t if

$$x_o(t) = -x_o(-t)$$



Even and Odd Function

Even and odd functions have the following properties:

- Even x Odd = Odd
- Odd x Odd = Even
- Even x Even = Even

Every signal $x(t)$ can be expressed as a sum of even and odd components because:

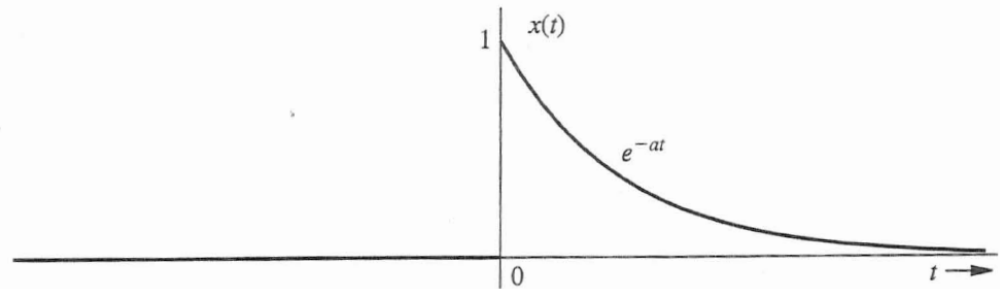
$$x(t) = \underbrace{\frac{1}{2}[x(t) + x(-t)]}_{\text{even}} + \underbrace{\frac{1}{2}[x(t) - x(-t)]}_{\text{odd}}$$

Even and Odd Function

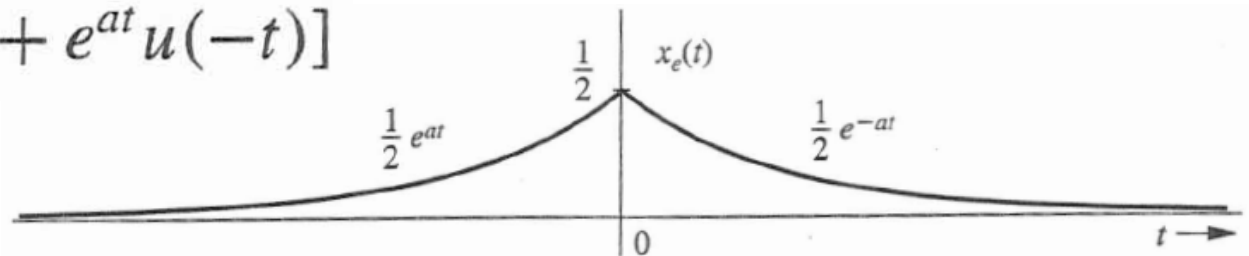
$$x(t) = \underbrace{\frac{1}{2}[x(t) + x(-t)]}_{\text{even}} + \underbrace{\frac{1}{2}[x(t) - x(-t)]}_{\text{odd}}$$

Example: Consider the causal exponential function $x(t) = e^{-at}u(t)$

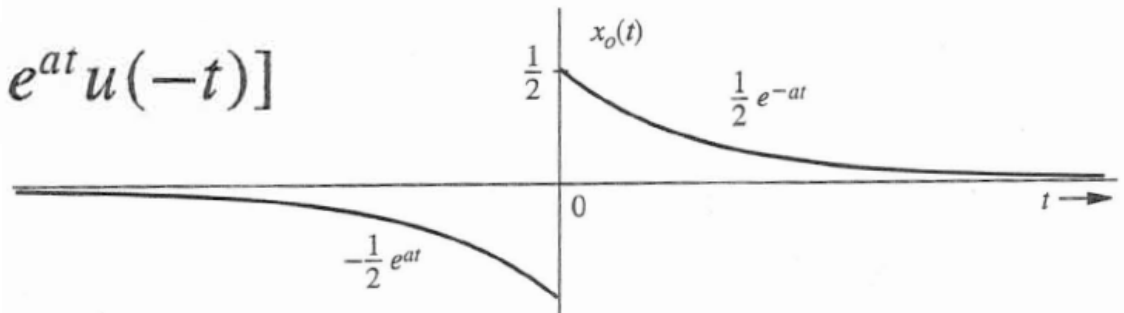
$$x(t) = x_e(t) + x_o(t)$$



$$x_e(t) = \frac{1}{2}[e^{-at}u(t) + e^{at}u(-t)]$$



$$x_o(t) = \frac{1}{2}[e^{-at}u(t) - e^{at}u(-t)]$$



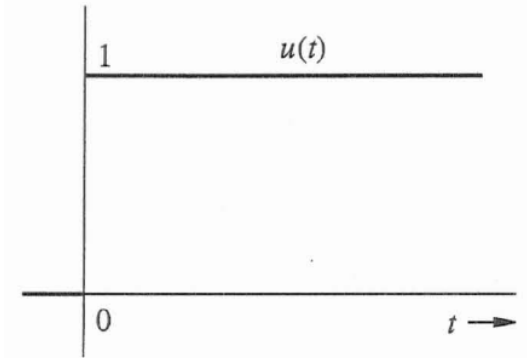
Signal Models

- Unit Step Function $u(t)$
- Pulse Signal $\Pi\left(\frac{t}{\tau}\right)$
- Unit Impulse Function $\delta(t)$
- Exponential Function e^{st}

Unit Step Function $u(t)$

Step function defined by:

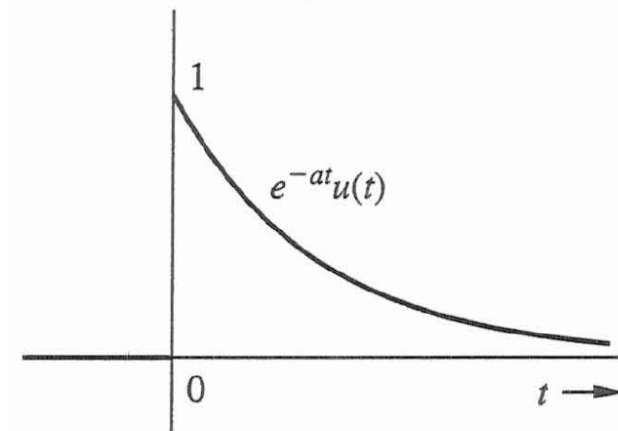
$$u(t) = \begin{cases} 1 & t \geq 0 \\ 0 & t < 0 \end{cases}$$



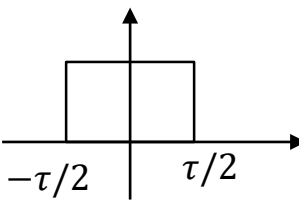
Useful to describe a signal that begins at $t = 0$ (i.e. causal signal).

For example, the signal e^{-at} represents an everlasting exponential that starts at $t = -\infty$.

The causal form of this exponential $e^{-at}u(t)$

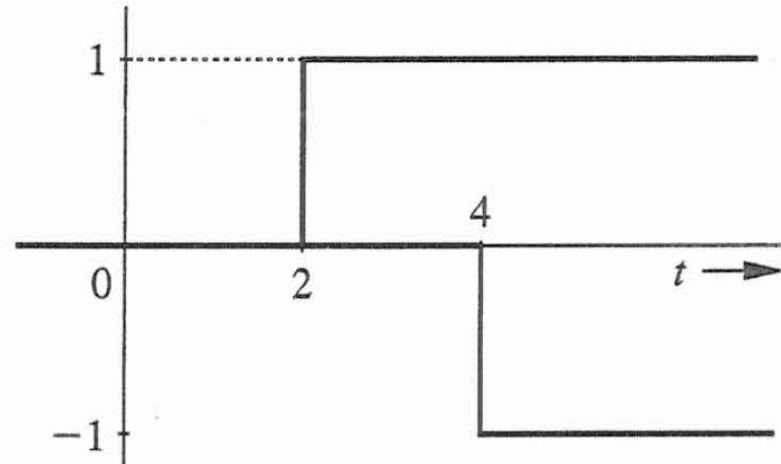
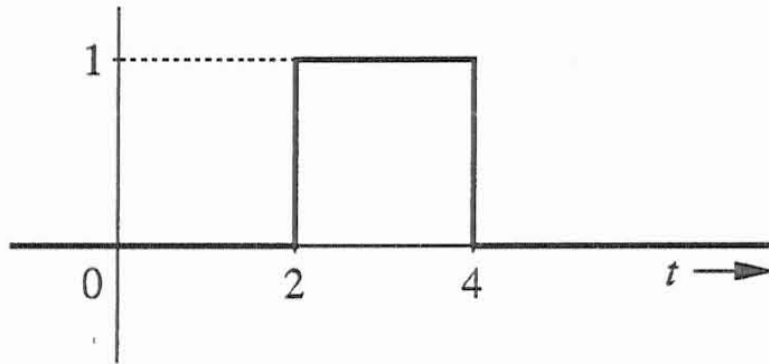


Pulse Signal

$$x(t) = \Pi\left(\frac{t}{\tau}\right)$$


A pulse signal can be presented by two step functions:

$$x(t) = u(t-2) - u(t-4)$$



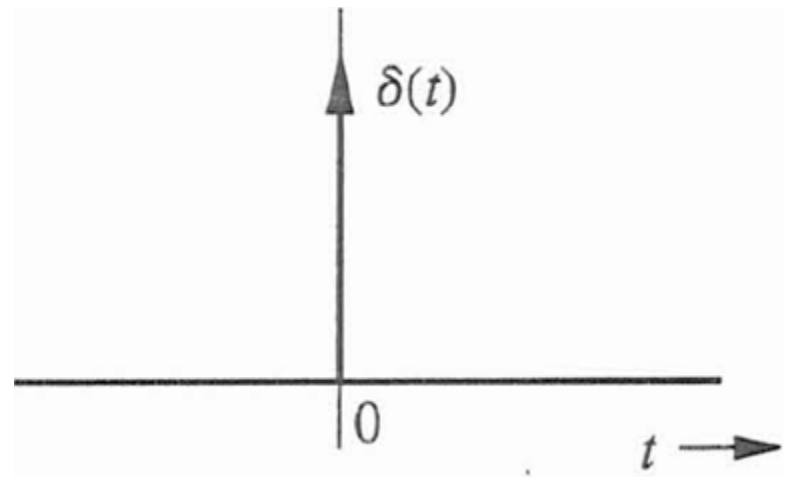
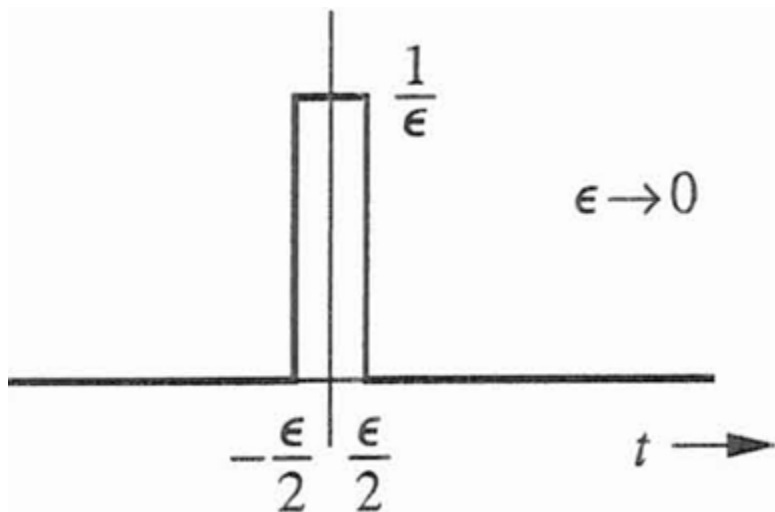
Unit Impulse Function $\delta(t)$

First defined by Dirac as:

$$\delta(t) = 0 \quad t \neq 0$$

$$\int_{-\infty}^{\infty} \delta(t) dt = 1$$

$$\frac{du(t)}{dt} = \delta(t)$$



Multiplying Function $\phi(t)$ by an Impulse

Since impulse is non-zero only at $t = 0$, and $\phi(t)$ at $t = 0$ is $\phi(0)$, we get:

$$\phi(t)\delta(t) = \phi(0)\delta(t)$$

We can generalize this for $t = T$:

$$\phi(t)\delta(t - T) = \phi(T)\delta(t - T)$$

Sampling Property of Unit Impulse Function

Since we have: $\phi(t)\delta(t) = \phi(0)\delta(t)$

It follows that:

$$\int_{-\infty}^{\infty} \phi(t)\delta(t)dt = \phi(0) \int_{-\infty}^{\infty} \delta(t)dt = \phi(0)$$

This is the same as “sampling” $\phi(t)$ at $t = 0$.

If we want to sample $\phi(t)$ at $t = T$, we just multiple $\phi(t)$ with $\delta(t - T)$

$$\int_{-\infty}^{\infty} \phi(t)\delta(t - T)dt = \phi(T)$$

This is called the “sampling or sifting property” of the impulse.

Examples

Simplify the following expression

$$\left(\frac{1}{j\omega + 2} \right) \delta(\omega + 3)$$

Evaluate the following

$$\int_{-\infty}^{\infty} \delta(t + 3) e^{-t} dt$$

Find dx/dt for the following signal

$$x(t) = u(t-2) - 3u(t-4)$$

The Exponential Function e^{st}

- Important in signal and system analysis.
- s is a complex variable (complex frequency)
 $s = \sigma + j\omega$.
- σ is the **decay rate** and ω is the **oscillation rate**.

Example

$$e^{st} = e^{(\sigma+j\omega)t} = e^{\sigma t} e^{j\omega t} = e^{\sigma t} (\cos \omega t + j \sin \omega t)$$

$$e^{s^*t} = e^{(\sigma-j\omega)t} = e^{\sigma t} e^{-j\omega t} = e^{\sigma t} (\cos \omega t - j \sin \omega t)$$

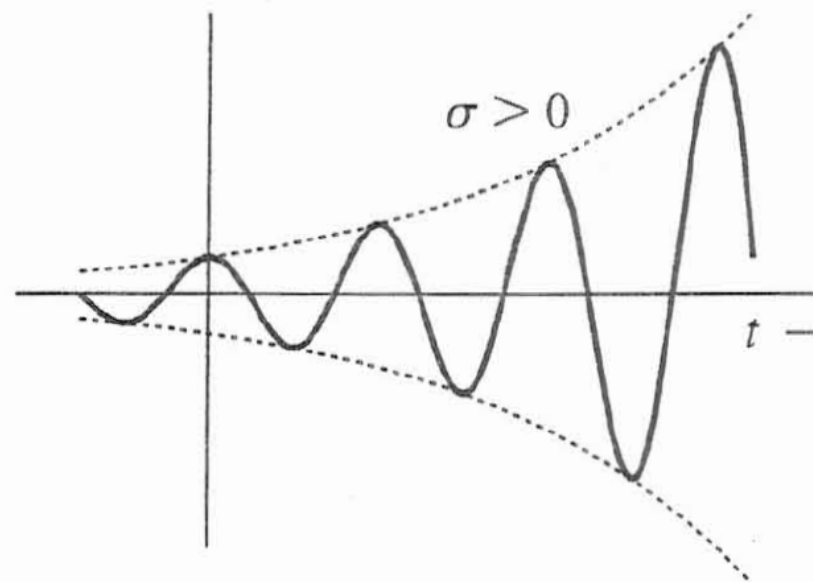
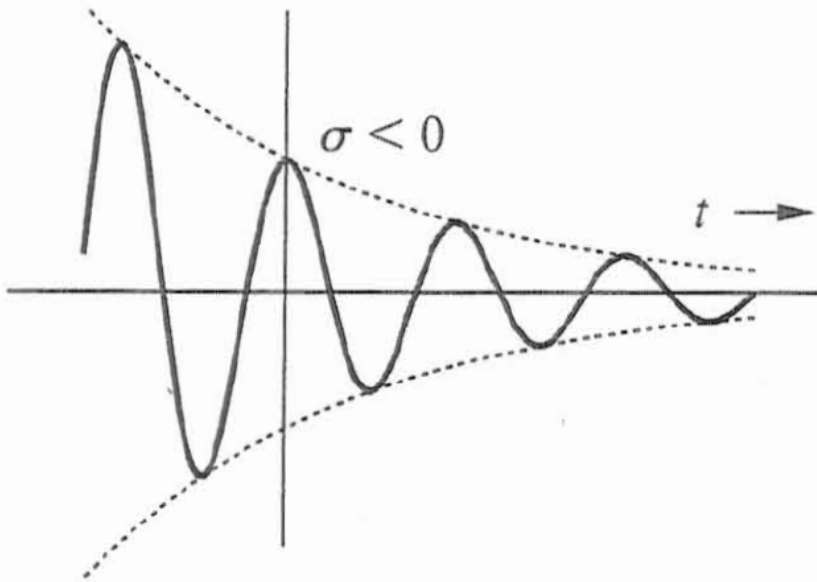
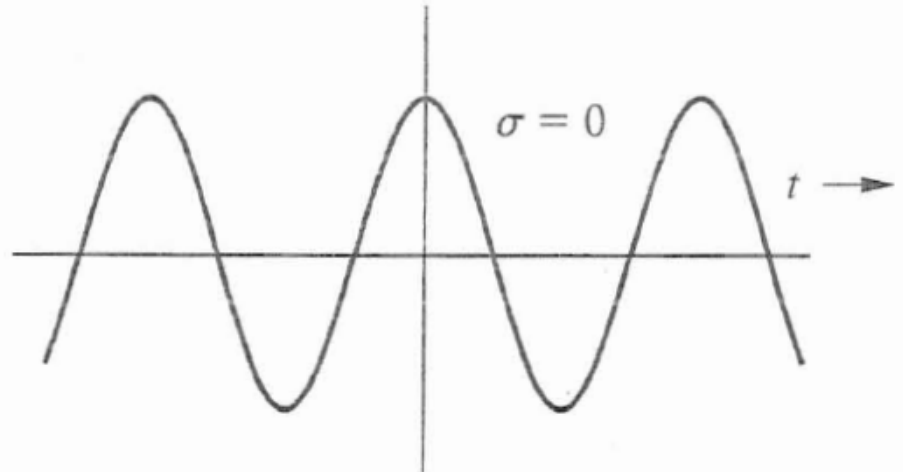
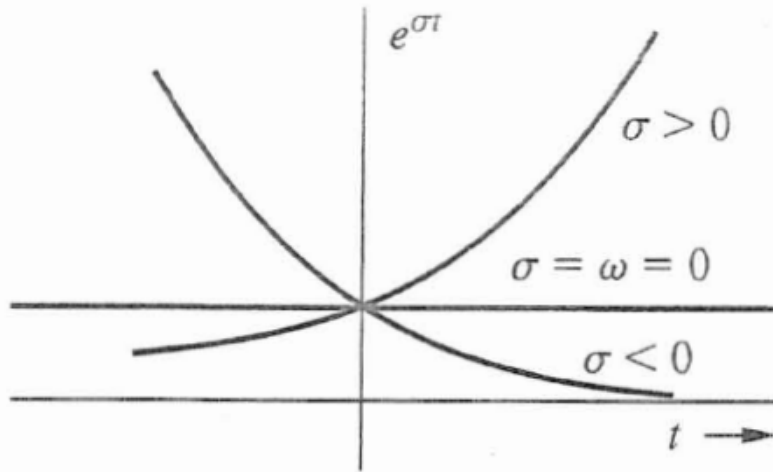
$$\frac{1}{2} (e^{st} + e^{s^*t}) = e^{\sigma t} \cos \omega t$$

The Exponential Function e^{st}

The function e^{st} can be used to describe a very large class of signals and functions.

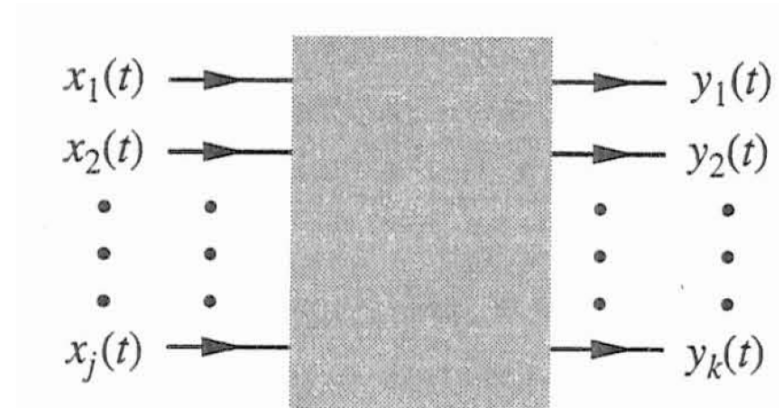
- | | | |
|-------------------------------|---|--------------|
| 1- A constant k | $x(t) = ke^{st} = ke^{0t} = k$ | $s = 0$ |
| 2- Exponential $e^{\sigma t}$ | $x(t) = e^{(\sigma + j\omega)t} = e^{\sigma t}$ | $\omega = 0$ |
| 3- Sinusoidal $\cos \omega t$ | $x(t) = \frac{1}{2} (e^{st} + e^{s^*t}) = e^{\sigma t} \cos \omega t$ | |
| | $x(t) = \cos \omega t$ | $\sigma = 0$ |

The Exponential Function e^{st}



What are Systems?

- Systems are used to **process signals to modify or extract information**
- Physical system – characterized by their **input-output relationships**
- E.g. Electrical systems are characterized by voltage-current relationships
- E.g. Mechanical systems are characterized by force-displacement relationships
- From this, we derive a **mathematical model of the system**
- **“Black box”** model of a system:



Classification of Systems

Systems may be classified into:

1. Linear and non-linear systems
2. Constant parameter and time-varying-parameter systems
3. Instantaneous (memoryless) and dynamic (with memory) systems
4. Causal and non-causal systems
5. Continuous-time and discrete-time systems
6. Analog and digital systems
7. Invertible and noninvertible systems
8. Stable and unstable systems

Linear Systems

- A **linear system** exhibits the **additivity property**:

$$\text{if } x_1 \rightarrow y_1 \text{ and } x_2 \rightarrow y_2 \text{ then } x_1 + x_2 \rightarrow y_1 + y_2$$

- It also must satisfy the **homogeneity or scaling property**:

$$\text{if } x \rightarrow y \text{ then } kx \rightarrow ky$$

- These can be combined into the property of **superposition**:

$$\text{if } x_1 \rightarrow y_1 \text{ and } x_2 \rightarrow y_2 \text{ then } k_1 x_1 + k_2 x_2 \rightarrow k_1 y_1 + k_2 y_2$$

- A non-linear system is one that is NOT linear (i.e. does not obey the principle of superposition)

Examples

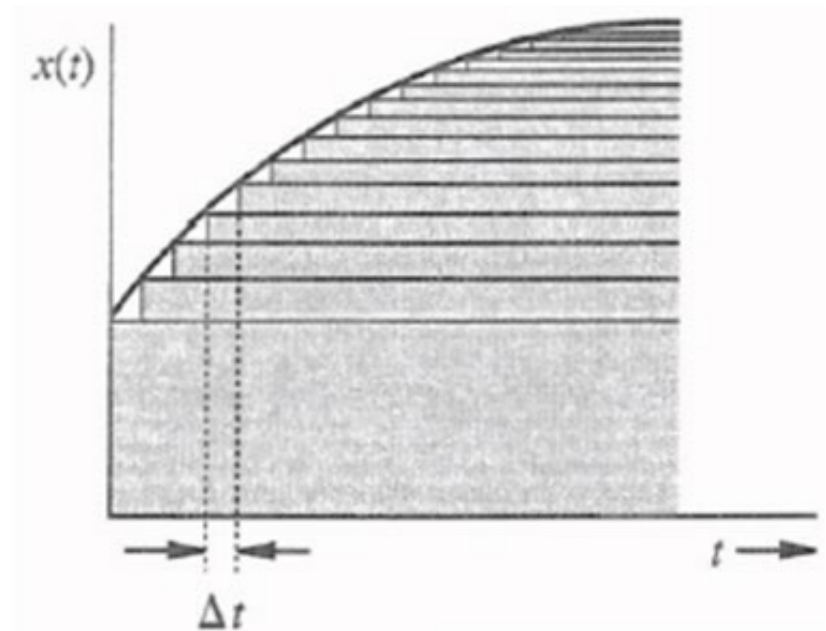
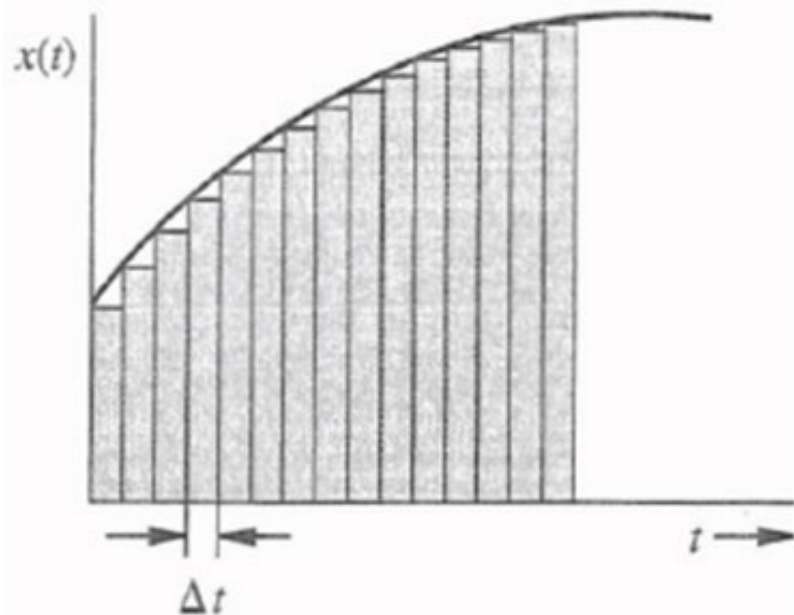
Determine if the system linear or non-linear

a) $\frac{dy}{dt} + 2y = x$

b) $y = x^2$

Advantage of Linear Systems

A complex input can be represented as a sum of simpler inputs (pulse, step, sinusoidal), and then use linearity to find the response to this simple inputs to find the system output to the complex input.

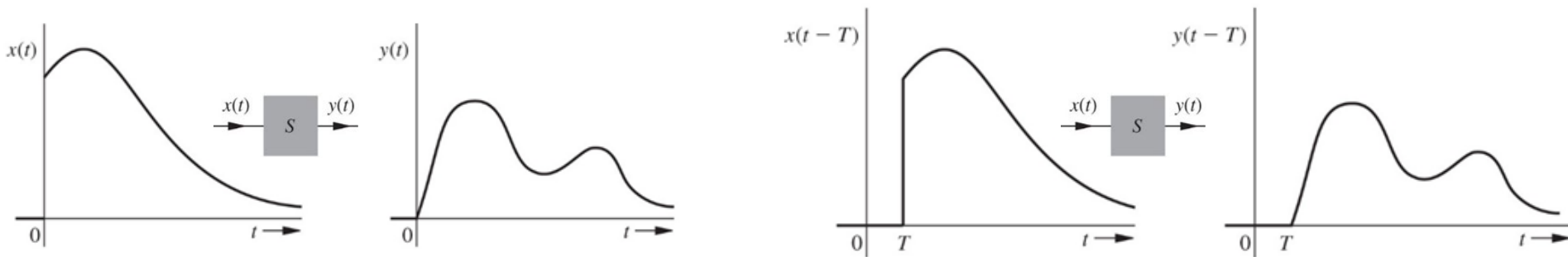


$$x(t) = a_1 x_1(t) + a_2 x_2(t) + \cdots + a_m x_m(t)$$

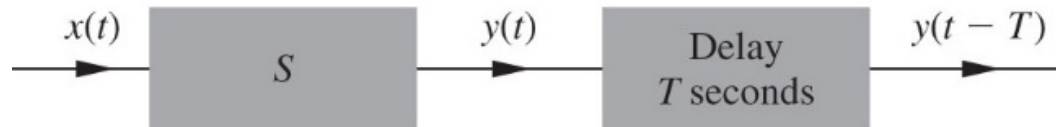
$$y(t) = a_1 y_1(t) + a_2 y_2(t) + \cdots + a_m y_m(t)$$

Time-Invariant System

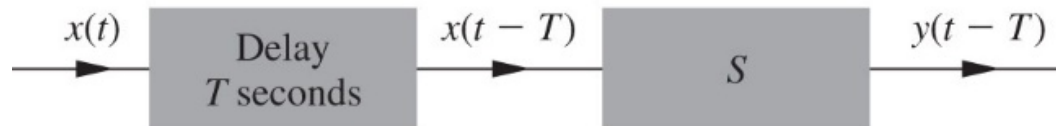
Time-Invariant system is a system whose parameters and response do not change with time.



Method to test time-invariant



(a)



(b)

Example

Determine if the system is time-invariant?

(a) $y(t) = 3x(t)$

(b) $y(t) = t x(t)$

Instantaneous and Dynamic Systems

Dynamic System: system's output at time t depends on the current input and past input (**system with memory**).

Instantaneous System: system's output at time t depends only on the current input. (**memoryless system**)

Which of the two systems is instantaneous?

a) $y(t) = 3 x(t)$

b) $y(t) = 3 x(t) + x(t-1)$

Causal and Noncausal Systems

Causal System: the output at any time instant t_0 depends only on the input $x(t)$ for $t \leq t_0$.

Present output depends on the past and present inputs, not on future inputs.

All practical real time system must be causal system since it cannot predict future input and produce an output based on future input.

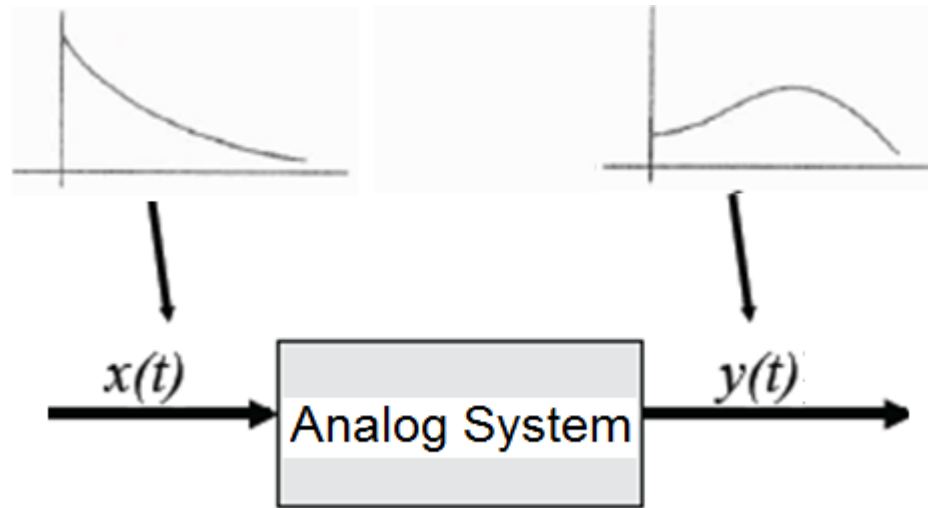
Which of the two systems is causal?

a) $y(t) = 3x(t) + x(t-2)$

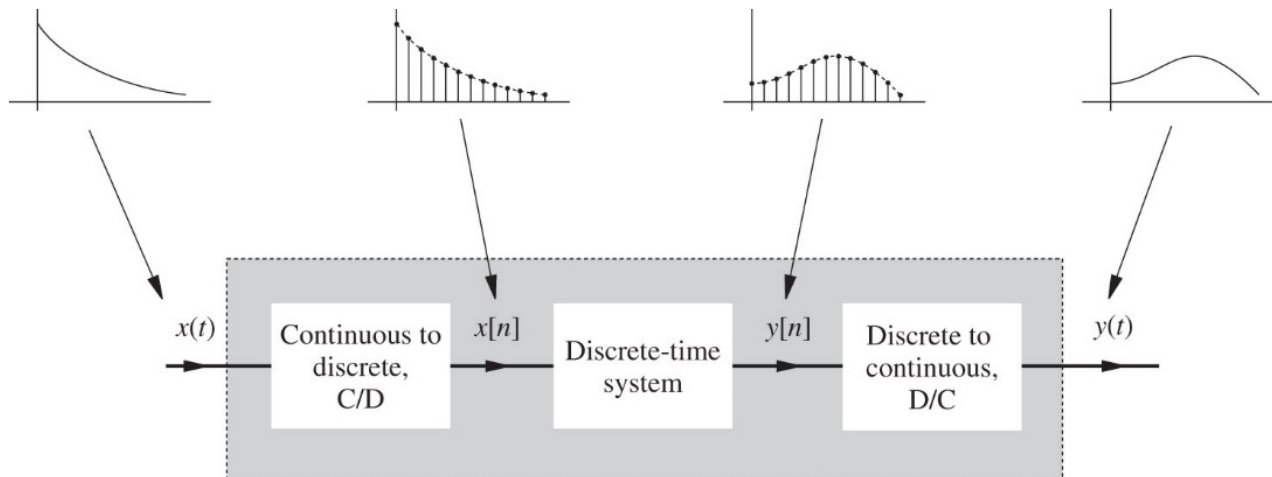
b) $y(t) = 3x(t) + x(t+2)$

Analog and Digital Systems

Analog System: Input is continuous and the output is continuous

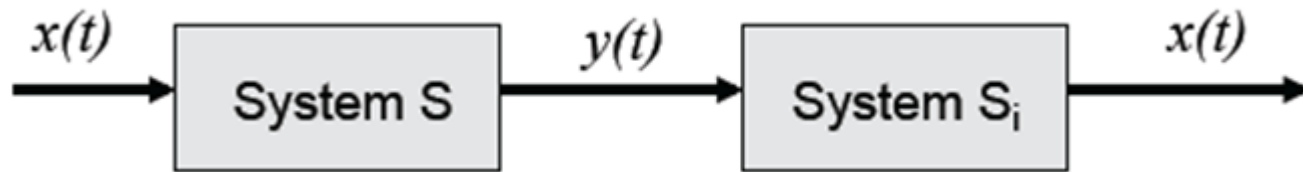


Digital System: Input is discrete and the output is discrete



Invertible and Noninvertible

- Let S_1 be a system whose output is $y(t)$ for input $x(t)$.
- S_1 is invertible if it is possible to design a system S_2 that takes the signal $y(t)$ as an input and produces an output that is $x(t)$. S_2 is the inverse of S_1 .
- System S_1 is invertible if it produces a unique output for every unique input, one to one mapping of the inputs to the outputs.



Which of the two systems is invertible?

a) $y(t) = x^2$

b) $y = 2x$

System External Stability (BIBO)

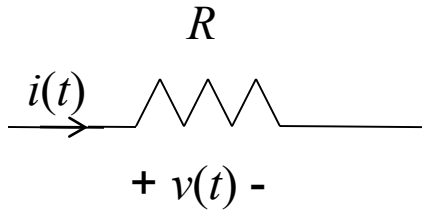
System is externally stable if for **b**ounded **i**ntermediate input it gives **b**ounded **o**utput.

System Model

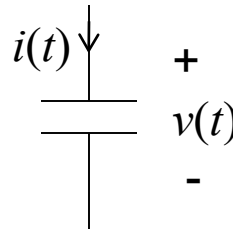
Many biological, electrical, and mechanical system can be modeled by a differential equation that relates the input $x(t)$ to the output $y(t)$.

The next task is to solve the differential equation to find the output $y(t)$ for specific input $x(t)$.

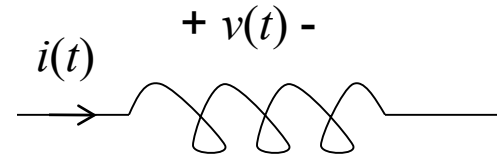
Electrical System



$$v(t) = R i(t)$$



$$i(t) = C \frac{dv}{dt}$$



$$v(t) = L \frac{di}{dt}$$

v : Voltage

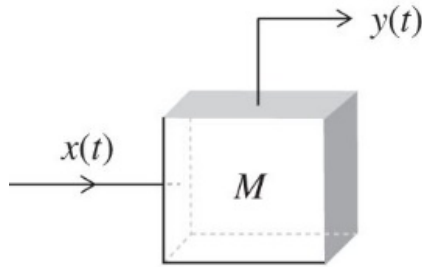
i : Current

R : Resistor

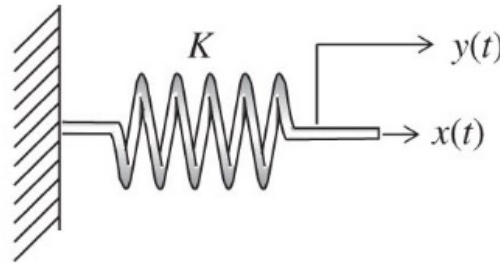
C : Capacitor

L : Inductor

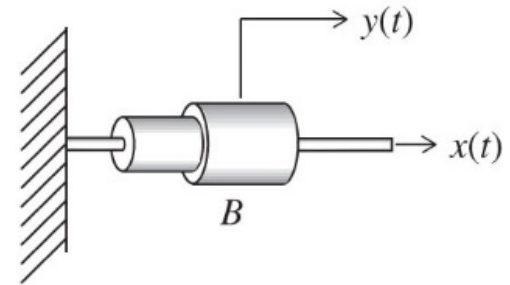
Mechanical System



(a)



(b)



(c)

$$x(t) = M \ddot{y}(t) = M \frac{d^2 y}{dt^2}$$

$$x(t) = k y(t)$$

$$x(t) = B \dot{y}(t) = B \frac{dy}{dt}$$

M : Mass

x : Force

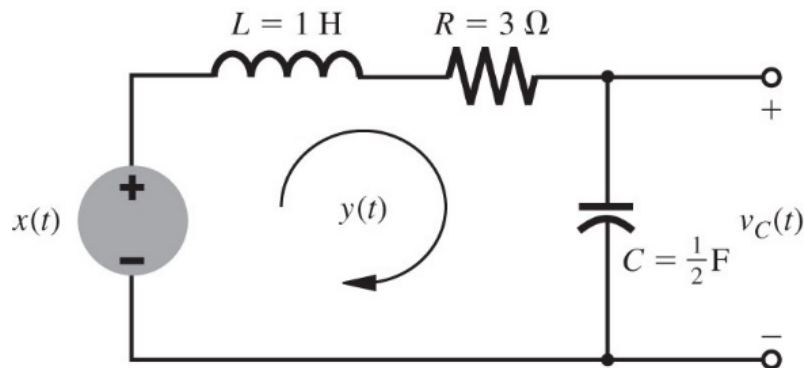
y : Displacement

k : stiffness constant of the spring

B : Damping coefficient of the dashpot

Example

Find the input-output relationship for the electrical system shown below. The input is the voltage $x(t)$, and the output is the current $y(t)$.



$$V_L + V_R + V_C = x(t)$$

$$L \frac{dy}{dt} + Ry(t) + \frac{1}{C} \int y dt = x(t)$$

$$\frac{d^2 y}{dt^2} + \frac{R}{L} \frac{dy}{dt} + \frac{1}{LC} y = \frac{1}{L} \frac{dx}{dt}$$

$$\frac{d}{dt} = D \qquad \frac{d^2}{dt^2} = D^2$$

$$D^2 y + \frac{R}{L} Dy + \frac{1}{LC} y = \frac{1}{L} Dx$$

$$D^2 y + 3Dy + 2y = Dx$$

$$(D^2 + 3D + 2)y = (D)x$$

↑
Characteristic Polynomial

Example 2

Find the input-output relationship for the transitional mechanical system shown below. The input is the force $x(t)$, and the output is the mass position $y(t)$.

$$x(t) - F_S - F_{DP} = Ma$$

$$x(t) - ky(t) - B\dot{y}(t) = M\ddot{y}(t)$$

